

27 *Use of the Data Booklet is relevant to this question.*

Electric cars use zinc-air battery with one electrode made up of zinc and the other is a carbon electrode. During discharge, zinc dissolves as ions. At the carbon electrode, hydroxide ions are formed. Zinc ions then react with the hydroxide ions, forming solid zinc hydroxide.

Which statements about this cell are correct?

- 1 The e.m.f of the cell is +1.16V.
- 2 The overall cell reaction is $2\text{Zn} + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Zn}(\text{OH})_2$.
- 3 Overall pH of the solution does not change at the end of the reaction.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only